24.2.1 Exercises - Production and use of X-rays

1.

a. Explain how the hardness of an X-ray beam is controlled by the accelerating voltage in the X -ray tube.

.....

.....[2]

b. The attenuation of a parallel beam of X-ray radiation is given by the expression

$$\frac{I}{I_0} = e^{-\mu x}$$

where μ is the linear attenuation (absorption) coefficient and x is the thickness of the material through which the beam passes.

i. State

a) what is meant by attenuation,

.....[1]

b) why the expression applies only to a parallel beam.

.....

.....[2]

ii. The linear attenuation coefficients for X-rays in bone and in soft tissue are 2.9 cm^{-1} and 0.95 cm^{-1} respectively.

Calculate, for a parallel X-ray beam, the ratio

$$\frac{\text{fraction } \frac{I}{I_0} \text{ of intensity transmitted through bone of thickness 2.5 cm}}{\text{fraction } \frac{I}{I_0} \text{ of intensity transmitted through soft tissue of thickness 6.0 cm}}.$$

ratio = ... [2]

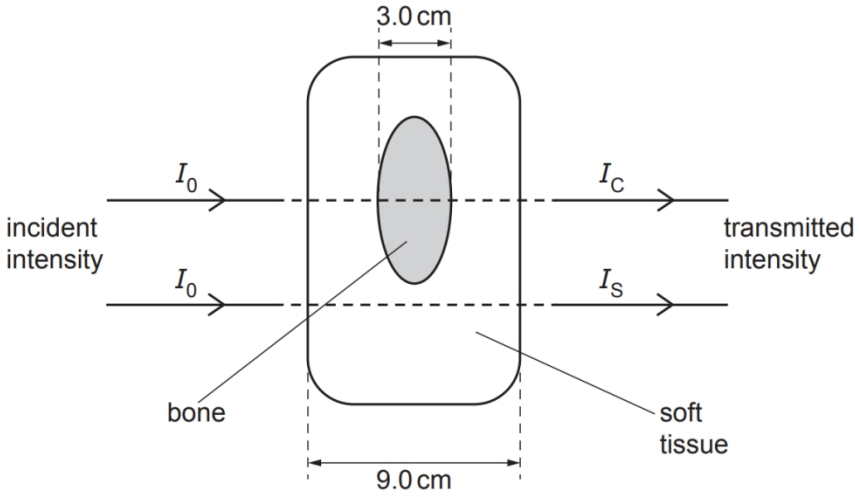
2.

a. State how, in a modern X-ray tube, the intensity of the X-ray beam and its hardness are controlled.

intensity: .....

hardness: .....[2]

b. A model of a limb consists of soft tissue and bone, as illustrated in Fig.



	μ/cm^{-1}
soft tissue	0.92
bone	2.9

The soft tissue has a thickness of 9.0 cm. The bone within the soft tissue has a thickness of 3.0 cm. Data for the linear attenuation (absorption) coefficient μ of X-rays in soft tissue and in bone are shown in Table.

A parallel beam of X-rays of intensity I_0 is incident normally on the model.

Calculate, in terms of I_0:

i. the transmitted intensity I_S through soft tissue alone

I_S = ... I_0 [2]

ii. the transmitted intensity I_C through soft tissue and bone.

I_C = ... I_0 [2]

c. By reference to your answers in (b), suggest, with a reason, whether good contrast on an X-ray image would be obtained.

.....[1]

3.

a. Electrons are accelerated through a potential difference of 15kV. The electrons collide with a metal target and a spectrum of X-rays is produced.

i. Explain why a continuous spectrum of energies of X-ray photons is produced.

.....

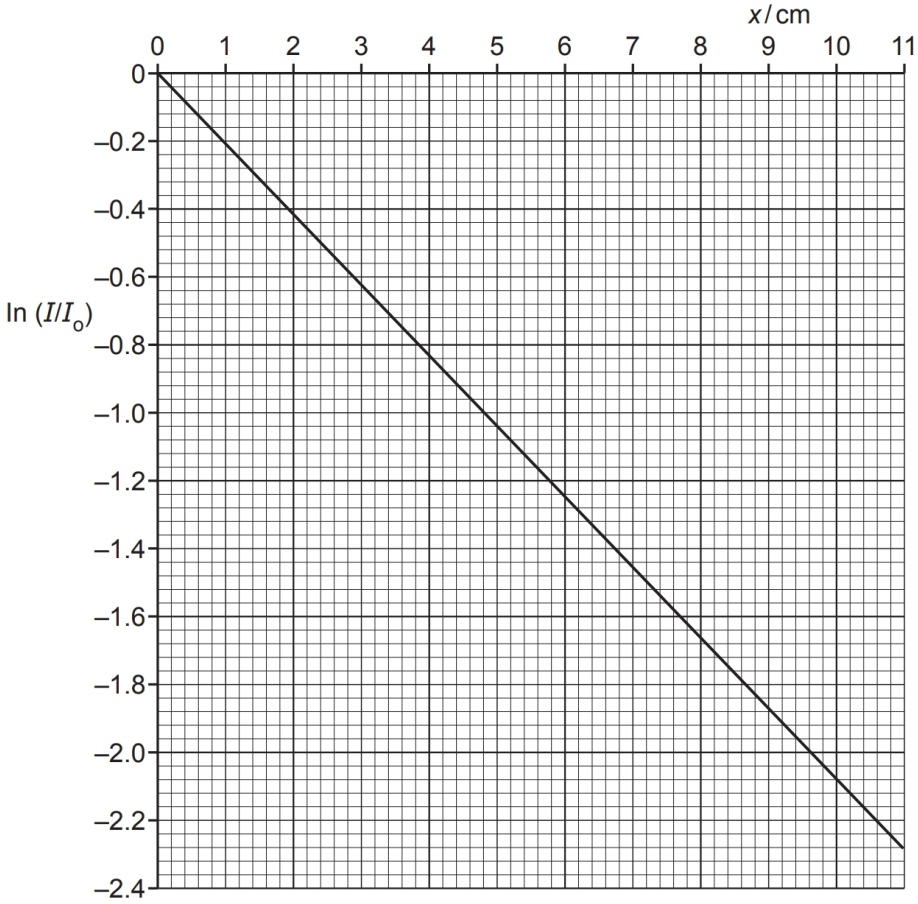
.....

.....[3]

ii. Calculate the wavelength of the highest energy X-ray photon produced.

wavelength = ... m [3]

b. A beam of X-rays has an initial intensity Io. The beam is directed into some body tissue. After passing through a thickness x of tissue the intensity is I. The graph in Fig. shows the variation with x of ln (I/Io).



i. Determine the linear attenuation (absorption) coefficient μ for this beam of X-rays in the tissue.

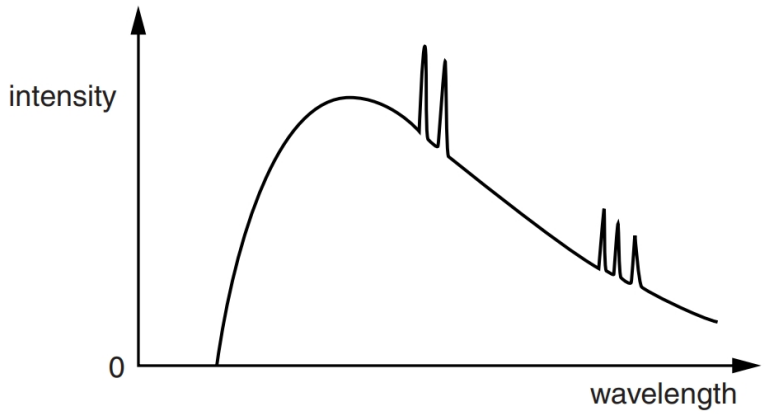
μ = ... cm^{-1} [2]

ii. Determine the thickness of tissue that the X-ray beam must pass through so that the intensity of the beam is reduced to 5.0% of its initial value.

thickness = ... cm [2]

4. High-energy electrons collide with a metal target, producing X-ray photons.

The variation with wavelength of the intensity of the X-ray beam is illustrated in Fig.



a. Explain why there is

i. a continuous distribution of wavelengths,

.....

.....

.....[3]

ii. a sharp cut-off at short wavelength,

.....

.....[2]

iii. a series of peaks superimposed on the continuous distribution of wavelengths.

.....[1]

b. In the X-ray imaging of body structures, longer wavelength photons are frequently filtered out of the X-ray beam.

i. State how this filtering is achieved.

.....[1]

ii. Suggest the reason for this filtering.

.....[1]